Archives of Dermatological Research (2025) 317:540
https://doi.org/10.1007/s00403-025-04022-3

ORIGINAL PAPER

Efficacy of intralesional acyclovir versus quadrivalent human papillomavirus vaccine for treatment of recalcitrant cutaneous warts: a clinical trial

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Received: 26 December 2024 / Revised: 2 February 2025 / Accepted: 12 February 2025
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Abstract

Warts are the cutaneous manifestations of human papilloma virus infection. Immunotherapy with human papillomavirus (HPV) vaccines has been tried with promising outcomes. Acyclovir is an antiviral with established efficacy against DNA viruses, could become a possible revolutionary therapeutic option for warts. This study aims to assess the efficacy and safety of intralesional acyclovir versus quadrivalent human papilloma virus vaccine in treatment of recalcitrant cutaneous warts. A total of 60 patients with recalcitrant warts were assigned into 3 groups: group I, 20 patients received Intralesional injection of acyclovir, group II, 20 patient received intralesional quadrivalent HPV vaccine, and group III, 20 patients received intralesional injection of normal saline. The patients were followed up monthly for 3 months after the last injection to detect any recurrence. Complete clearance was observed in 8 patients (40%) in group I, 16 patients (80%) in group II and no response in group III. The most common side effects were pain (100%) among all groups, hemorrhagic eschars in group I(100%), edema in group II(30%) while the least was flu like symptoms (5%) in group II. both intralesional quadrivalent HPV vaccine and intralesional acyclovir seem to be promising, well-tolerated therapeutic options for the treatment of warts, with statistically significant superiority of the quadrivalent vaccine over acyclovir.

Keywords Wart · Acyclovir · Human papillomavirus · Vaccine

Introduction

Cutaneous and anogenital warts are clinical manifestations of an infection with human papillomaviruses. Cutaneous warts can affect any part of the skin, but they mainly appear on the fingers, soles, and face. Common warts come in various sizes and present as firm, hyperkeratotic, rough papules that can form large masses through confluence on the dorsum of hands and fingers, as well as around or beneath the nails (periungual warts) [1].

A subset of HPV types 16, 18, 26, 31, 33, 35, 39, 45, 51, 53, 56, 58, 59, 62, 66, 68, 73, and 82 have been designated as high-risk types [2]. In common warts, HPV 1, 2, 4, and 7 are the most commonly associated genotypes. These warts are prevalent in 3.5% of adults and 33% of children [3].

Transmission occurs through direct skin-to-skin contact or indirectly through contaminated surfaces and objects, such as swimming pools and towels. Autoinoculation of the virus is also a significant factor in the spread of the virus [1].

Several modalities have been used in the treatment of viral warts, including.

destructive methods such as cryotherapy, electrodesiccation and curettage, topical salicylic acid, and immune-based therapies. Traditional destructive measures act locally on the infected tissue and can be associated with pain, scarring, and high recurrence rates [4]. Furthermore, photodynamic therapy (PDT), is proving to be a valuable therapeutic option for the treatment viral warts, especially if multiple and in difficult-to-treat areas such as the acral ones [5].

Immune-based therapies, such as intralesional immunotherapy, have been tried using a variety of agents such as Candida antigen, mumps antigen, combined measles,

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Published online: 08 March 2025

mumps, and rubella (MMR) vaccine, tuberculin purified protein derivative, and Bacille Calmette-Guerin vaccine [6].

Intralesional injections of vaccines have been extensively studied in the treatment of warts, showing some degree of success. Although the mechanism is not fully understood, these injections are believed to work by inducing a systemic T-cell mediated response [7].

The quadrivalent HPV vaccine targets HPV16, 18, 6, and 11. HPV6 and 11 are responsible for approximately 90% of genital warts [8].

Intralesional injection of HPV vaccines has been shown to be more effective in treating cutaneous warts compared to the standard intramuscular administration [9].

There have been few studies and case reports that previously reported the use of oral and topical acyclovir in the treatment of warts, some of which were resistant to previous treatments [10–14]. The effectiveness of intralesional acyclovir injection in treating common warts was assessed by Elsayed et al. in 2022, and 52.6% of their patients experienced a complete response, suggesting it as a potential treatment option [15].

Acyclovir is a synthetic analogue of a purine nucleoside with antiviral properties against certain viruses such as varicella-zoster and herpes simplex. The drug is metabolized into its active form, acyclovir triphosphate, by the virus thymidine kinase enzyme. This enzyme targets viral DNA to prevent viral replication within the host [16].

The suppression of HPV by acyclovir can be explained by possible phosphorylation of the drug by cellular enzymes. This process inhibits the integration of viral DNA and the replication of cellular DNA in infected keratinocytes. Acyclovir acts as both an antiviral and an antineoplastic drug. Additionally, it may induce apoptosis of the HPV-infected cells [10].

Based on the aforementioned results, we conducted this study to assess the efficacy of intralesional acyclovir versus quadrivalent human papillomavirus vaccine in the treatment of recalcitrant cutaneous warts aiming to address the gap not covered by previous studies on acyclovir. This study is the first trial to compare the effectiveness of intralesional acyclovir with intralesional quadrivalent HPV vaccine as an immunotherapeutic agent in treating recalcitrant cutaneous warts.

Patients and methods

This is a three arm, single blinded, randomized controlled clinical trial included 60 cases with common warts. The study was performed under the principles of the Declaration of Helsinki [17] The study was approved by the institutional review board of Damietta faculty of medicine, Al-Azhar University (DFM-IRB00012367-24-01-002). A written informed consent was obtained from all patients.

Patients were divided into 3 groups: Group I included 20 patients received intralesional injection of acyclovir in concentration 70 mg/ml. (Aciclovir MYLAN 250 mg) with a dose of 0.1 ml injected into each wart at the base with a total dose 1 ml per session. Group II, included 20 patients received intralesional injection of 0.1 ml of quadrivalent HPV vaccine into the base of the largest wart, the vaccine manufactured by Merk Sharp &Dohme Corp., US, a subsidiary of Merck &Co., Inc. Group III, included 20 patients received intralesional injection of normal saline 0.9% as a control at a dose of 0.3 ml. The injection was repeated every two weeks until complete clearance of warts or for a maximum of five sessions. Clinical and dermoscopic photographs were taken before treatment and at each follow-up visit. The photographs have been evaluated objectively by 2 blinded dermatologists. The response to treatment was categorized by the regression in the size of warts into 3 grades: complete response (100% total resolution of warts), partial response (50–99% resolution of warts) and no response (less than 50% resolution of warts). Adverse effects were reported, and the patients were followed up monthly for 3 months after the last injection to detect any recurrence.

Patients received other treatment modalities within the last month, Patient with >10 warts, Pregnant, lactating women, history of hypersensitivity, patients who are immunocompromised, with active TB or history of TB, or on immunosuppressive drugs in addition to acute febrile diseases or autoimmune disorders were all excluded from the study.

All patients were subjected to full history including personal history, history of present illness, including disease duration and previous treatment for warts. History of systemic or other dermatological diseases and history of medications (corticosteroids or immunosuppressant). Furthermore general, dermatologic and dermoscopic examinations were conducted for all patients included in the study.

Statistical analysis

Statistical analysis was performed with SPSS statistical software, version 25 (IBM, Chicago, Illinois, USA). The normality of the data will be tested by the Shapiro Wilk test. Qualitative data was presented as numbers and percentages and was compared by the Chi square /Fisher's exact test, while quantitative data was presented as mean and standard deviations/ median and interquartile range and was compared by the independent t test/ Mann Whitney U test, which is appropriate according to normality of data. As a result, the p-value was considered significant at the level of <0.05.

Results

In the current study, the median age in group I was 32 years, in group II it was 26 years, and in group III it was 32 years. There was no statistically significant difference between the groups in terms of age ($P=0.114$).

Regarding sex distribution, 25% of Group I were male and 75% were female, while in Group II 50% were male and 50% were female, and in Group III 35% were male and 65% were female. There was no statistically significant difference between the groups in terms of sex distribution ($P=0.535$).

According to the type of warts, the most common type of wart in groups I and III were common warts, while in group II it was plantar warts. Additionally, there was a statistically significant difference in distant warts among the three studied groups, with a p-value of 0.036. The highest percentage was in group II at 65%. Furthermore, the hand and foot were the most common sites of warts, with percentages of 40%, 40%, and 50% for each group, and 55%, 60%, and 50% respectively. There was no statistically significant difference between the studied groups regarding the site of the disease ($P=0.847$).

Regarding the number of warts, the median number of warts in group I was 4, in group II it was 4.10, and in group III it was 4.45. There was no statistically significant difference between the cases in the different studied groups regarding the number of warts ($P=0.661$).

The median disease duration in Group I was 2.75 years. In Group II, the median disease duration was 3 years. Group III had a median disease duration of 2.95 years. There was no statistically significant difference between the cases in the different studied groups regarding the disease duration ($P=0.596$).

Regarding previous treatments, ablative therapy was the most common in group II (50%), while topical keratolytic was the most common in group I. Systemic treatment was the least common (5.0%) Table 1.

Table 2 demonstrates a highly statistically significant difference between the three studied groups in treatment response, with a p-value of less than 0.001. Group II showed the highest treatment response, with complete resolution at 80%, compared to only 40% in group I (acyclovir). Additionally, while partial response in group I was 50%, it was only 20% in group II. Two patients (10%) did not respond to treatment in group I (Figs 1–5).

Regarding the response to distant warts within the three studied groups, no response was detected in group I. In group II, a complete response was detected in 8 out of 13 patients (61.5%), a partial response was found in 3 out of 13 patients (23%), and no response was observed in 15.5% of patients. In group III, no distant partial or complete local

Table 1 Baseline characteristics of the studied patients

	Group I (Acyclovir) ($N=20$)	Group II (Quadrivalent HPV vaccine) ($N=20$)	Group III (Control group) ($N=20$)	*P* value
	Mean	Mean	Mean	
Age	32 (19–45)	26 (18–43)	32 (21–40)	0.055
Sex				
Male	5 (25.0%)	10 (50.0%)	7 (35.0%)	0.535
Female	15 (75.0%)	10 (50.0%)	13 (65.0%)	
Disease duration(years)	2.75 (1–3)	3 (2–4)	2.95 (3–5)	0.596
Number of warts	4 (2–8)	4.10 (3–9)	4.45 (3–6)	0.661
Distant wart	8 (40.0%)	13 (65.0%)	5 (25.0%)	0.036
Type of wart				
Common	9 (45.0%)	7 (35.0%)	10 (50.0%)	0.932
Palmar	2 (10.0%)	4 (20.0%)	3 (15.0%)	
Planter	7 (35.0%)	8 (40.0%)	6 (30.0%)	
Periungual	2 (10.0%)	1 (5.0%)	1 (5.0%)	
Site of wart				
Hand	8 (40.0%)	8 (40.0%)	10 (50.0%)	0.847
Foot	11 (55.5%)	12 (60.0%)	10 (50.0%)	
Abdomen	1 (5.0%)	0(0.0%)	0(0.0%)	
Size (cm)				
< 0.5	10 (50.0%)	14 (70.0%)	10 (50.0%)	0.338
> 0.5	10 (50.0%)	6 (30.0%)	10 (50.0%)	
Previous treatment				
Topical salicylic acid	11 (55.0%)	7 (35.0%)	6 (30.0%)	0.439
Ablative therapy	7(35.0%)	10 (50.0%)	9(45.0%)	
Immunotherapy	2 (10.0%)	2 (10.0%)	3 (15.0%)	
Systemic treatment(acitretin)	0(0.0%)	1(5.0%)	0 (0.0%)	

Table 2 Treatment response among studied cases

Treatment response	**Group I (Acyclovir)**		**Group II (Quadrivalent HPV vaccine)**		**Group III (Control group)**		***P* value**
	Count	**%**	**Count**	**%**	**Count**	**%**	
Complete response (100%)	8	40.0%	16	80.0%	0	0.0%	<0.001
Partial response (50–99%)	10	50.0%	4	20.0%	0	0.0%	
No response (<50%)	2	10.0%	0	0.0%	20	100.0%	
Distant response	**Group I (Acyclovir) *N*=8**		**Group II (Quadrivalent HPV vaccine) *N*=13**		**Group III (Control group) *N*=5**		**P value**
Complete response (100%)	0	0.0%	8	61.5%	0	0.0%	<0.001
Partial response (50–99%)	0	0.0%	3	23.0%	0	0.0%	
No response (<50%)	8	100.0%	2	15.5%	5	100.0%	

response was detected. There was a statistically significant difference in the response between the three studied groups, with the highest response seen in group II Table 2.

Table 3 shows that pain during injection was reported in all groups. A statistically significant difference in itching, with a p-value of 0.030, was observed among the studied groups. The highest value of itching was seen in group II at 20%. Additionally, there was a statistically significant difference in edema, with a p-value of 0.005. Group I had the highest percentage of edema at 40%. Moreover, there was a highly statistically significant difference in hemorrhagic eschar among the studied groups, with the highest



Fig. 1 Male patient 42 years old with common wart before (**A**) and after (**B**) treatment with intralesional quadrivalent HPV vaccine

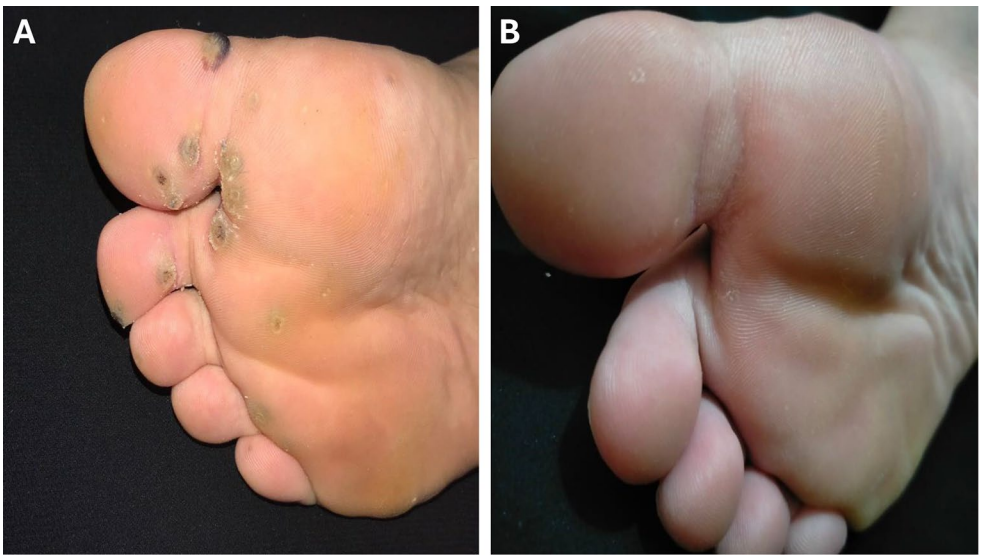


Fig. 2 Female patient 19 years old with planter wart before (**A**) &after (**B**) treatment with intralesional quadrivalent HPV vaccine



Fig. 3 Female patient 43 years old with common wart before (**A**) and after (**B**) treatment with intralesional quadrivalent HPV vaccine



Fig. 4 Female patient 33 years old with common wart before (**A**) and after (**B**) treatment with intralesional acyclovir

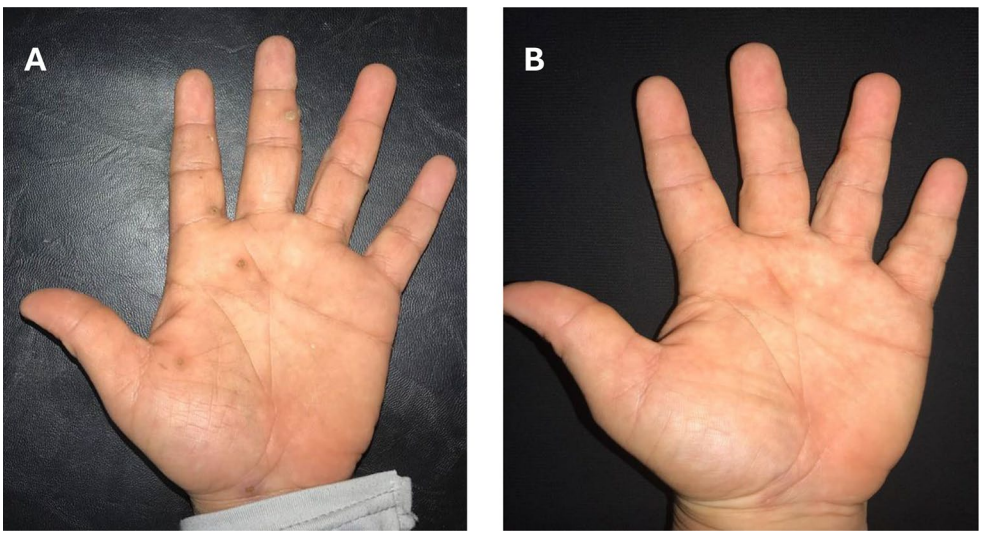


Fig. 5 Female patient 35 years old with palmer wart before (**A**) and after (**B**) treatment with intralesional acyclovir

Table 3 Adverse effects among studied cases

Adverse effects	Group I (Acyclovir)		Group II (Quadrivalent HPV vaccine)		Group III (Control group)		*P* value
	Count	%	Count	%	Count	%	
Pain during injection	20	100.0%	20	100.0%	17	85.0%	0.100
Itching	0	0.0%	4	20.0%	0	0.0%	0.030
Edema	8	40.0%	6	30.0%	0	0.0%	0.005
Erythema	3	15.0%	4	20.0%	2	10.0%	0.900
Flu like symptoms	0	0.0%	1	5.0%	0	0.0%	1
Hemorrhagic eschar	20	100.0%	0	0.0%	0	0.0%	<0.001
Recurrence	2	10.0%	0	0.0%	0	0.0%	0.322

percentage in the acyclovir group. Furthermore, recurrence was observed in 10% of the acyclovir group.

Discussion

Warts are common, benign, epidermal lesions caused by HPV, which infects epithelial tissues of the skin and mucous membranes [18].

Approximately 65% of warts clear spontaneously within 2 years, however 35% tend to be recalcitrant either showing no response to treatment or having prompt recurrences after treatment. No single treatment is completely effective in all patients and the success rate of both destructive and immunological procedures varies from 65 to 85% [19].

Acyclovir has antiviral action against certain viruses, such as varicella-zoster and herpes simplex, by targeting viral DNA to inhibit replication within the host. Intralesional acyclovir is a newly utilized treatment modality [20].

Human papillomavirus (HPV) vaccines, administered either intramuscularly or intralesionally, have shown promise in the treatment of common warts [9].

Our study illustrated that ablative therapy was the most common previous treatment in group II (50%), while topical keratolytic was the most common in group I. Moreover, laser and systemic treatments were the least common. Elsayed et al. (2022) reported a history of previous wart treatments in 17 patients (54.8%), including cryotherapy, trichloroacetic acid, and laser therapy [15].

In the present study, the most common type of warts in group I and group III were common warts, while in group II it was plantar warts. Furthermore, hands and feet were the most common sites for warts. Moreover, the highest percentage of distant warts was found in group II (65%).

Our findings are consistent with a study conducted by Meghana Reddy et al. (2023), which found that in the acyclovir group, six (30%) participants had a common wart, four (20%) had a palmar wart, and six (30%) had a plantar wart. In the intralesional PPD group, seven (35%) had a common wart, five (25%) had a palmar wart, and four (20%) had a plantar wart [20].

According to the treatment response in this study, there was a highly statistically significant difference between the three studied groups in treatment response (p-value < 0.001). Moreover, group II (quadrivalent HPV vaccine) showed the highest treatment response with a complete resolution rate of 80%, while the complete response in group I (acyclovir) was 40%.

Tandeter and Tandeter (2005) reported complete resolution of resistant plantar warts after oral intake of valacyclovir 1 gm for coexistent herpes zoster [14]. Similarly, Bagwell et al. (2016) described a case of recalcitrant plantar warts that disappeared 10 days after acyclovir therapy for herpes zoster [12].

In 2022, Elsayed et al. evaluated the efficacy and safety of intralesional acyclovir in the treatment of thirty-one patients with cutaneous warts. They achieved complete clearance of warts in 10 patients (52.6%), while seven patients had a partial response (36.8%), and three patients showed no response to therapy (10.5%). A highly statistically significant difference was found between the treatment and control groups. The researchers concluded that acyclovir can be an effective and well-tolerated treatment modality for cutaneous warts [15].

The present study aligns with a study by Meghana Reddy et al. (2023) that compared intralesional acyclovir versus immunotherapy for the treatment of viral warts. They found that in the group treated with intralesional acyclovir and the group treated with intralesional PPD, six (30%) and three (15%) participants reported a partial response at two weeks, respectively. A complete response was observed in 12 (60%) and six (30%) participants at 12 weeks, respectively. The difference in the proportion of therapeutic response between the two treatment modalities was shown to be significant in the fourth week (P value < 0.05). In the acyclovir group, cases of common warts showed a complete response in three patients (25%) in the 12th week. In the case of palmar warts and plantar warts, a complete response was observed in four (33.3%) participants each, whereas in periungual warts and subungual warts, only one (8.33%) participant each showed a complete response [20].

In contrast, Gibson et al. (1984) compared liquid nitrogen with acyclovir cream and placebo cream for the treatment of

plantar warts. They reported that there were no beneficial effects of acyclovir cream over placebo cream for the treatment of plantar warts [21].

The present study aligns with Gharib et al., who compared intralesional acyclovir with hepatitis B vaccine. They found that there was no statistically significant difference between the HBV vaccine group and the Acyclovir group. However, both groups showed a statistically more powerful response compared to the control group, which was injected with saline. Out of the 16 patients in the three studied groups, 7 (43.8%), 6 (37.5%), and 3 (18.7%) respectively achieved a complete response [22].

In the study conducted by Fouda et al. in 2024, they compared the use of intralesional quadrivalent Human Papillomavirus vaccine versus candida antigen in the treatment of multiple recalcitrant Non-Genital Warts. The study found that the highest response rate was seen in the HPV vaccine group, followed by the candida antigen group. In the candida antigen group, a complete response was observed in 33.3% of cases, with a partial response in 50%. In the quadrivalent HPV vaccine group, a complete response was seen in 72.7% of cases, with a partial response in 27.3%. These findings support our results regarding the effectiveness of the HPV vaccine [23].

Kumar et al. (2021) reported a complete clearance rate of 83.3% in 10 out of 12 patients with genital warts after receiving intralesional injections of the quadrivalent HPV vaccine [24].

The present study is in agreement with Nofal et al. (2023), who compared the efficacy and tolerability of intralesional quadrivalent vaccine versus intralesional bivalent vaccine in the treatment of multiple recalcitrant warts. A total of 50 patients with various types of multiple recalcitrant warts were randomly assigned to three groups: group I (quadrivalent vaccine, 20 patients), group II (bivalent vaccine, 20 patients), and group III (saline control, 10 patients). The study found a statistically significant difference in the therapeutic response between the quadrivalent vaccine group and the other two groups [25].

Regarding the distant response rate, the highest complete response rate was detected in the HPV vaccine group at 61.5%, while the partial response rate was 23%. There were no responses in the acyclovir and control groups.

These results are in agreement with a study conducted by Fouda et al. (2024), who concluded that in terms of the distant response rate, the highest response rate was detected in the HPV vaccine group, followed by the candida antigen group. In the candida antigen group, a complete response was detected in 33.3% and a partial response in 50%, while in the quadrivalent HPV vaccine group, a complete response was detected in 72.7% and a partial response in 27.3% [23].

Regarding side effects, the most common side effect in this study was pain during injection, which was observed in all groups. Erythema was reported in 15% of group I, 20% of group II, and 10% of group III. There was a statistically significant difference in edema (p value, 0.005), with group I having the highest percentage (40%). Itching also showed a statistically significant difference (p value, 0.03) among the studied groups, with the highest percentage (20%) in group II. Additionally, there was a highly statistically significant difference in hemorrhagic eschar among the studied groups, with the highest percentage in group (I) The least common side effect was flu-like symptoms (5%) in group (II) Recurrence was reported in 10% of group I.

Elsayed et al. (2022) discovered that adverse effects of intralesional injection of acyclovir included pain during injection in 17 patients (89.5%), localized reactions at the site of injection in the form of erythema in a single patient (5.3%), and blistering in 10 patients (52.6%) [15].

Furthermore, in a study conducted by Meghana Reddy et al. (2023), it was found that 20 out of 20 participants (100%) reported a burning sensation, three out of 20 participants (15%) reported desquamation, and 18 out of 20 participants (90%) reported pain in the acyclovir-treated group. In contrast, in the PPD-treated group, 12 out of 20 participants (60%) had no side effects, while eight out of 20 participants (40%) reported pain.

The present study agrees with Nofal et al. (2023) who found that adverse effects were mild, transient, and tolerable with both vaccines. These effects were similar to those reported with regular, highly safe intramuscular HPV vaccination programs [25].

In addition, Fouda et al. (2024) reported that the side effects of intralesional quadrivalent HPV vaccine in the treatment of non-genital recalcitrant warts were erythema and fever in 25% and 10% of cases, respectively. Tolerable pain during injection was reported in all the included cases [24].

The mechanisms underlying the therapeutic role of HPV vaccines in treating warts are not fully understood. These mechanisms include the induction of a long-term neutralizing humoral immune response against the targeted HPV types, as well as a lesser extent of a cellular immune response. Additionally, the vaccines enhance virus recognition by cell-mediated immunity and promote the production of cross-reactive antibodies [26].

Acyclovir's proven effectiveness against DNA viruses makes it a highly recommended antiviral treatment option for warts [27].

To the best of our knowledge, this is the first randomized controlled study comparing the efficacy of the intralesional quadrivalent HPV vaccine as an immunotherapeutic agent to intralesional acyclovir, an antiviral drug, in treating

recalcitrant warts. However, the study had some limitations as it was conducted at a single center and recruited a relatively small sample size. Future studies are recommended to explore the benefits of acyclovir with a larger sample size, compare it to other conventional therapies, and investigate other routes of administration as well.

Conclusion

In the treatment of viral warts, the therapeutic response shown by intralesional acyclovir is promising, with no severe or life-threatening side effects. However, we have concluded that the efficacy of intralesional quadrivalent HPV vaccine is superior to that of intralesional acyclovir.

Author contributions D.Mona ibrahim elyamany: making the practical part of the study, writting the main manuscript, revision. D.zakaria mahran Obaid: revision. D.Ibrahim fouda: corresponding author, prepared figures, revision, submission all authors discussed the results and contributed to the final manuscript.

Funding None.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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